

SceneSound Studio Pro - Final Report

CS 5588 Week 15 Final Hands-On Challenge

Executive Summary

SceneSound Studio Pro is a Week 15 final multimodal AI system that builds directly on Week 13 controlled animal-care image generation and Week 14 SceneSound audio generation. The final system creates image prompts, soundtrack prompts, SFX cue sheets, narration scripts, responsible-AI reviews, and evaluation metrics from one structured scenario.

Technical Approach

The pipeline accepts animal, condition, environment, mood, duration, and audience inputs. It normalizes the inputs, creates image-generation prompts, creates music prompts, plans sound effects, writes a voice-ready narration, checks safety issues, and reports prompt alignment, realism, diversity, latency, safety flags, and cue counts.

Experiments and Evaluation

The evaluation compares the Week 14 audio-only baseline against the Week 15 multimodal final system. Metrics include prompt alignment, multimodal quality, realism, diversity, latency, safety flags, and SFX cue coverage. This makes progress measurable instead of only visual.

Business Value

Target users include veterinary educators, animal shelters, pet-care creators, instructors, and small marketing teams. The product reduces production time for educational media by combining visual direction, soundtrack planning, SFX, narration, and safety review in one workflow.

Responsible AI

The project addresses graphic medical content, misleading veterinary advice, copyright risks in music generation, and voice cloning misuse. A production version should label AI outputs, avoid diagnosis claims, require licensing checks, and restrict cloned voice use without consent.

Lessons Learned

The main lesson is that foundation-model projects become stronger when separate model capabilities are combined into a useful workflow. Week 15 improves engineering, UX, evaluation, business framing, and safety while keeping the same multimodal creative direction.

Conclusion

SceneSound Studio Pro is significantly better than Week 14 because it turns a sound prompt prototype into a full presentation-ready multimodal AI production studio connected to both Week 13 and Week 14.

